Using VLMs to simulate subjective human perception and evolve interesting robots designs

reformulate for precise contribution

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Abstract. Designing robots with expressively interesting morphologies and behaviors typically requires hand-crafted reward functions, a laborious process fundamentally limited when the target is subjective. We investigate whether a Vision-Language Model (VLM) can replace this manual specification by acting as a zero-shot reward model in a Quality-Diversity evolutionary loop, reproducing the "subjective interestingness evaluation" of a robot by a human against. A VLM grader is prompted to score MuJoCo-simulated robots across two strategies : $(\mu+\lambda)$ and MAP-Elites. The VLM evaluate morphology via static image-based grading, which produces coherent fitness signals (Pearson correlation of 0.5 with human evaluators). We extend the approach to video-based grading for behavioral search. However the latter approach faces challenges, only capturing movement outlines and suffering from spatial hallucinations. Our results establish VLM-based grading as a viable reward model for simple subjective robotic design, while identifying limitations such as spatial reasoning and scoring inconsistency that motivate future work.

Keywords: Vision-Language Model · Robotic · Evolutionary Algorithm

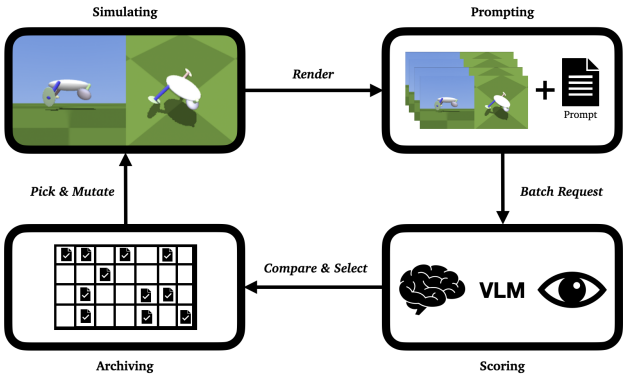


Fig. 1: Overivew of the proposed framework. A VLM is used to evaluate proposed robot designs based on a natural language description of a subjective target.

1 Introduction

Quality-Diversity algorithms, and MAP-Elites offer a compelling framework for difficult problems where diverse high-performing solutions are searched in a very wide space. Their power, however, depends on the quality of the performance criterion and the relevance of the feature descriptors used to populate the archive, both non-trivial to define when the target is inherently subjective.

Automatically designing robots is part of these problems : building morphologies and behaviors typically requires hand-crafted reward functions, a process that is both time-consuming and fundamentally limited when the target is related to abstract concept. Describing "walk efficiently" in terms of equations is tractable, describing "express sadness through motion" is considerably harder. Yet humans can make such judgments effortlessly, instantly recognizing whether a morphology or behavior resonates with a known concept, aesthetic, or emotion. Automatically reproducing this "human evaluation" skill is the purpose of this study : how to synthesize a "subjective evaluation of interestingness", a measure of how much a tested candidate feels "interesting" 3.1 for a target related to a known concept.

Recent advances in Vision-Language Models (VLMs) suggest a promising path forward for imitating human skills. Rocamonde *et al.*, [2] demonstrated that pre-trained VLMs can serve as zero-shot reward models for reinforcement learning in a robotic context, and Zhang *et al.*, [1] showed that foundation models can encode human notions of novelty and relevance as flexible evaluation signals. These works converge on the same core intuition: large pre-trained models already internalize a rich, human-like understanding of the world, and this understanding can be repurposed as a flexible evaluation signal.

This work asks whether that intuition can be extended to robotic morphology and motion. We investigate whether VLM image scoring can guide morphology search toward subjective concept such as "kangaroo" or "lamp". We extend the approach to VLM video scoring for controller search toward target behaviors. To address these questions, we implement a modular framework combining MuJoCo simulation, a generative morphology space, and a VLM evaluator (Gemini 3 Flash) that scores rendered images or videos against natural language prompts. We first benchmark the VLM as a grader, measuring human correlation, score guidance impact, and output consistency before applying it to morphology and controller search in two evolutionary modes: $(\mu + \lambda)$ and MAP-Elites.

The central objective is to establish whether a zero-shot VLM is able to reproduce the "subjective" human way of scoring a visual robot morphology and behavior according to a target to replace the human-in-the-loop in interactive evolutionary algorithms.

Our results establish VLM-based grading as a viable alternative to hand-crafted reward functions for subjective robotic design, achieving a Pearson correlation of 0.5 with human evaluators for morphology scoring. Behavior search revealed weaknesses of the video-based evaluation: the VLM captures only movement outlines and frequently produces spatial hallucinations. We further identify limitations in rendering sensitivity, VLM inconsistency and temporal reasoning.

2 Background

Automatic robotic design brings numerous challenges, such as morphological shaping or behavioral control, usually addressed through evolutionary robotics (J. R. Muff *et al.*, [4]). A central question is how to select the most promising individuals. Factual performance scores are typically used (Radhakrishna Prabhu SG *et al.*, [6]), but this approach is limited when the goal is complex or subjective (Costelloe D. *et al.*, [7]).

Human evaluators can fill this gap through interactive evolutionary computation, capturing subjective judgements that are hard to express mathematically (Eiben *et al.*, [12]). However, they are slow, inconsistent, and subject to fatigue.

Recent advances in artificial intelligence suggest an alternative. Deep learning has been used to simulate creative search for shapes and visuals (Nguyen *et al.*, [5]), and foundation models can mimic human judgment across many tasks (Zhang *et al.*, [1] and Elyoseph *et al.*, [8]). Large Vision-Language Models (VLMs) in particular reproduce both human vision and analysis (Zhang *et al.*, [9] and Zongxia Li *et al.*, [15]), making them candidates to replace the human evaluator as a reward model [14]. However, they remain limited in spatial reasoning (W. Hong *et al.*, [18] and Yu Songsong *et al.*, [19]).

Several works have used VLMs as zero-shot reward models. Rocamonde *et al.*, [2] optimized robotic controllers with reinforcement learning and showed the importance of rendering quality and model size, but only with factual targets. Kumar *et al.*, [13] used foundation models to search for interesting and emergent patterns in artificial life simulations, such as Lenia and Boids. We use a VLM in a similar role, simulating a human evaluator of robot morphologies and behaviors. Sam Earle *et al.*, [16] revisited the *picbreeder* experiment [20], replacing the human by a VLM.

Miyazaki *et al.*, [21] evolved animats with a genetic algorithm using VLM pairwise comparisons of images, on classic locomotion tasks. We tried with subjective targets and video-based evaluation for movement scoring . Qiu *et al.*, [22] used a Large Language Model to guide the mutation of robot morphologies. In their work the model drives the evolution, while we use it to score performance. We discuss combining both roles in our future work. In this work, the VLM is guided by score range on criteria to improve its performance, similar to Tao Shi *et al.*, [17] for industry product design.

Finally, Quality-Diversity algorithms can benefit from VLM-based evaluation. Novelty search has proven effective for robotic optimization (Mouret J. *Novelty-Based Multiobjectivization*, [10]), and the feature descriptors in MAP-Elites (Mouret J., Jeff C., [11]) are hard to define for subjective targets. VLMs allow abstract, language-based descriptors such as adjectives or subjective comparisons, opening a new search space.

3 Methodology

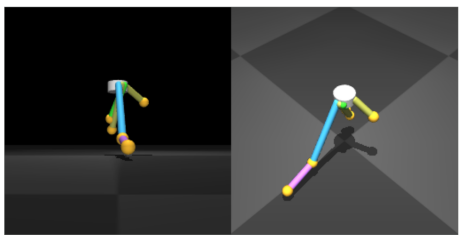
3.1 VLM as Reward Model

Known Limitations The first clear limitation is computation time. Inference takes a few seconds to perform, which is very long for an optimization loop.

The second issue is model consistency. VLMs are very sensitive to rendering [2] and may give very different answers depending on minor visual details. For example, in this study using a MuJoCo environment, a blue sky in the simulation can dramatically improve VLM performance compared to a default Grey or white background.

But the **major current limitation remains spatial reasoning**, as many key visual information cues are misunderstood or missed by the models.

Theses limitations are model-dependent and tend to diminish with newer and larger models. This study was conducted with **Gemini 3 Flash** (December 2025), chosen as a tradeoff between speed and reasoning ability. Using batched API requests of 10 individuals, we achieved evaluation rates of one image every 5s and one 5-second video every 6s, at approximately 12k and 16k tokens per batch respectively. For comparison, the lighter Gemini 3 Flash Lite is twice as fast but lacks sufficient spatial reasoning for most vision tasks, as shown figure 2.



Observation :[...]. It features four primary limb assemblies attached around the torso. The limbs appear to have two segments each,[...]. In the current pose, one blue-purple limb is extended toward the floor, while the other limbs are retracted or angled awkwardly.

Fig. 2: *Gemini 3 Flash Lite* description with hallucination. Sentences are color-coded by accuracy: green for correct, yellow for inaccurate, and orange for wrong. This lightweight model is fast but error-prone, as shown here by the incorrect assertion that all limbs have two segments when only one does. We have therefore used *Gemini 3 Flash* which produces significantly fewer errors.

Not to mention that the prompt plays a major role in outlining the context and guiding the task, but it may also artificially introduce unwanted bias. We focused on having a very general prompt explaining the global scoring task, into which the user target is injected. ***See website for the full prompts and variants used.***

The results reported in this paper are representative of equivalent models of the same generation (Qwen3-VL-30B-A3B, InternVL3.5-30B-A3B, GPT-5 mini), and are likely to be exceeded by newer models such as Gemini 3.5 Flash (May 2026).

Interestingness Evaluation The VLM is used as a proxy to reproduce human subjective perception, analysis, and evaluation. Given an image or video, the model is asked to describe it and evaluate its "interestingness" with respect to the target (figure 3). We used the VLM as zero-shot reward model with the defaults API parameters to evaluate its base performance.

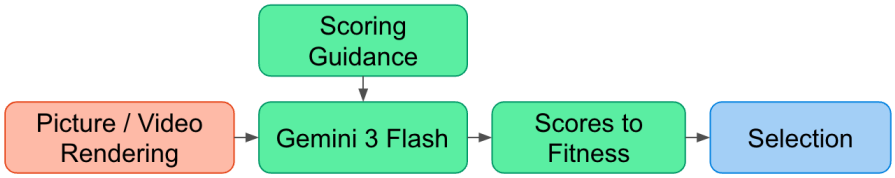


Fig. 3: VLM grading pipeline : VLM as reward models with a textual prompt and a batched set of images or video in input, and output a json formatted text containing the scores.

"**Interestingness**" is here defined as a linear combination of **coherence** with the target, **originality**, and **potential** for further improvement, each dimension $\in [0, 100]$. These three criteria are broken down into score ranges, each range having a short description of expectations and guidance for grading. An example of the scoring guidance for the coherence dimension:

```
coherence - How well does the morphology match the static target ?
0-29   = no recognizable similarity to a {target_morph}
30-49  = vague resemblance, very weak matching feature
50-69  = partial match, some features of {target_morph} are present
70-89  = some great resemblance, key features identifiable
90-100 = good match, structurally faithful to a {target_morph}
```

The **overall end fitness** is the weighted mean :

$$\text{fitness} = \frac{\text{coherence} + 0.5 \times \text{originality} + 1.5 \times \text{potential}}{3}$$

The use of a VLM allows the attribution of subjective descriptors for the MAP-Elites archive. Indeed, it is possible to assign properties of a subjective nature such as resonant concepts, objects resemblances, or specials expressions. Nevertheless, this work does not explore model-generated descriptors : all features are predefined like the target prompt by the human in natural language, and the VLM assigns the one that fits best.

An example of a feature descriptor for and elephant target :

```
How too characterized the elephant-like trunk ?
0-1 = no clear trunk identifiable
2-3 = weird trunk with clumsy shape (too much twisted, folded, locked)
4-5 = basic trunk-like shape, basic articulation
6-7 = great organic elephant-like trunk with great moving clearance
8-10 = a robotic shape trunk with special mechanism like gripper
```

To pursue the objective evoked in the introduction 1, we conducted different experiments focusing on three quantifiable variables : the score **correlation** between human and VLM (4.1), the score **guidance impact** (4.1), the VLM **consistency** in giving a trustworthy score (4.1).

To measure the correlation between the VLM and a human, we asked one human evaluator to grade morphologies according to a target on the three criteria. We then asked the VLM to grade the same morphologies and compared the results. We used the Pearson correlation coefficient to evaluate the linear correlation : it ranges from -1 (exact opposite linear correlation) to 1 (perfect linear match).

To verify the actual impact of the score ranges, we tested the VLM evaluation with and without score range guidance to measure its effect. To verify consistency, we made identical requests multiple times and measured the standard deviation of the scores given.

3.2 Robot Environment and Optimization

The robotic environment is based on Mujoco simulator (python library for realistic physics simulation). For rendering the scene, we chose a two side-view approach (figure 4), one from the side, and one from above at 45°. A green checkered floor has been added for considering distances. The images and the videos are in 720x360 pixels. The video were sampled at 5 frames per second during 5s, but the API automatically resample it at 1 frames per second (default VLM settings), resulting in 5 frames for movement evaluation. This low framerate is a tradeoff between optimization speed (order of magnitude is 1s of inference per second of video) and minimal information for understanding the scene.

Morphology Search The goal is to generate morphologies that fit a user target of a subjective nature, such as "a kangaroo able to jump high", "an ergonomic and stylish lamp", or "a crab with large claws". To do so, morphologies are generated randomly and mutated across an evolutionary process. A Vision-Language Model (i.e. Gemini 3 Flash) is then prompted to evaluate and score each morphology according to the user target. The score(s) given are used for elite selection and the evolutionary process.

The selection approach differs between $(\mu+\lambda)$ and MAP-Elites, in both cases the best fitness scores are kept, but with MAP-Elites, fitness scores are compared only between individuals in the same niche of specific features.

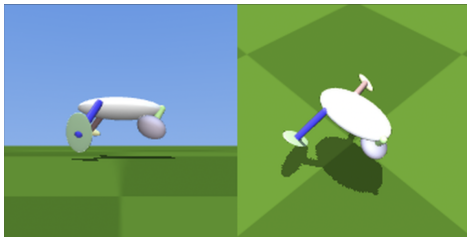


Fig. 4: Example of morphology reachable in the search space : The robot's morphology are composed of body parts (white ellipsoid) and legs (colored cylinder). The robot are mutated by adding/removing/modifying legs and body part according to gaussian probability. Applying the mutation operator many times allows to reach a large amount of possibilities.

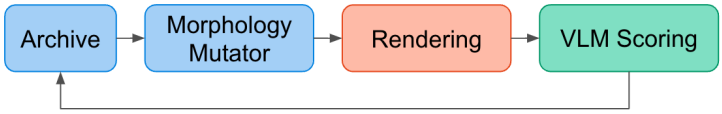


Fig. 5: **Morphology optimization pipeline.** Random morphologies are rendered and scored by the VLM, then selected to populate the archive. Saved individuals are then mutated iteratively to improve their scores. The two strategies : $(\mu + \lambda)$ selection and MAP-Elites, which differ in how the archive is structured and maintained.

The resources allocated for every run were limited due to computation time : in general, only 100–300 total individuals across 5–15 generations.

Controller Extension We extended the same approach to dynamic behavior: rather than evolving robot shapes, can a VLM guide the evolution of robot movements toward a subjective natural language target such as "a jumping robot" or "an upright rotating and dancing robot"?

The controller search introduces a second, inner optimization loop: robot movements are controlled by a neural network trained with PPO, whose objective is defined by a weighted combination of physical and interpretable terms, such as forward displacement, rotation, or jump height. The outer evolutionary loop searches over these weight combinations rather than directly in controller space, keeping the search space low-dimensional and meaningful: every combination produces an identifiable behavior and small mutations lead to predictable variations.

Each individual is evaluated three times independently, with the final score averaged to reduce VLM scoring inconsistency. The main limitation of this approach is that the vocabulary of physical terms is fixed, behaviors requiring incentives not present in the term library cannot be fully discovered.

More experimental details, results, and video examples are available on the accompanying website in the behavior section.

4 Results and Discussion

All the following experiment result can be reproduce with the available codebase[3]. We also provide a website containing more visual context and complete prompt example.[4]

4.1 VLM Benchmark

Correlation The experiment was run with different morphologies and targets[5]. As the figure 6 reports, **It eventually achieved a Pearson correlation around 0.5 for the fitness score, which confirms a real potential for simulating a human evaluator for morphology.** The gap to 1 may be explained by different factors : imperfect prompt, scoring bias or simply VLM inconsistency and scoring inability for complex concept.

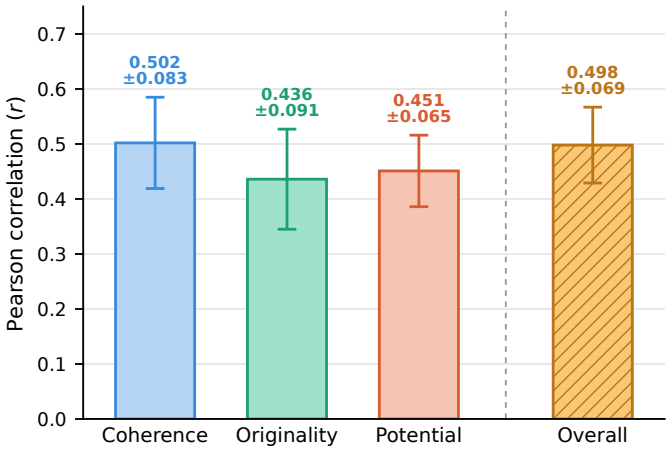


Fig. 6: synthesis of a 5 correlation evaluations on a dataset of 64 human evaluated morphologies : We asked one human evaluator to score the robots morphologies compared to specific targets. Then the VLM gave its score and we evaluate the correlation between the two scorers. It appears to reach a Pearson correlation around 0.5, verifying the existence of a clear correlation between the two scorers.

[3] https://anonymous.4open.science/r/VLM4ROBOT/README.md
[4] https://sites.google.com/view/vlm-4-interesting-robots
[5] dataset example available on the website (benchmark section).

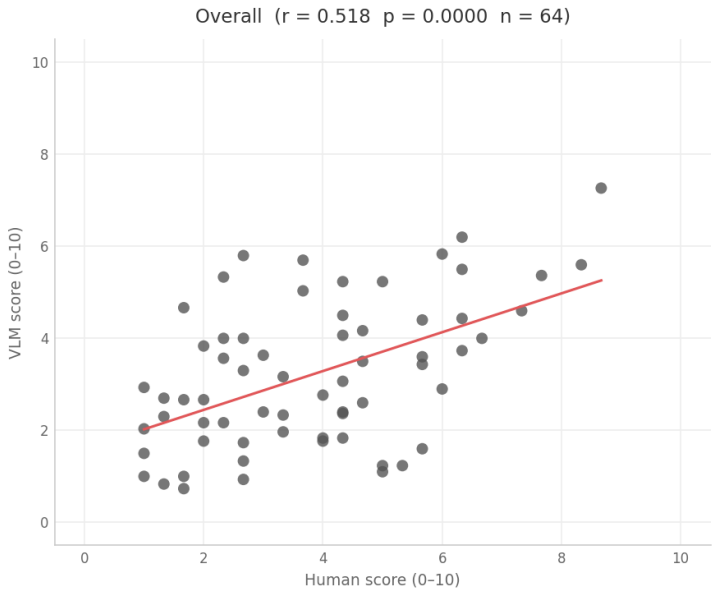


Fig. 7: Example of a correlation evaluations on a dataset of 64 human evaluated morphologies : We asked one human evaluator (x-axis) and the VLM (y-axis) to score the robots morphologies on several targets

The figure 7 gives an example of a run, highlighting the correlation with the slope of the regression of 0.5. Tuning the score guidelines in the prompt can influence this value : changing the score ranges and their associated descriptions impact the correlation. This is the approach used in the following applications. Nevertheless, this approach has multiple downsides: it is time-consuming, and not easy to tweak, as shown in figure 8.

Guidance via prompting

Figure 8 report the impact of the guidelines in the scoring process : for relatively good candidate, the scoring guidelines actually help pulling the VLM score toward the human score. However, when the morphology has a low human score, the guideline still tends to push the VLM to be too optimistic. Adjusting the lower part of the score ranges partially mitigates this issue. That said, it is more beneficial too pull the great score upward to make great solution selected even if it still pulls up a bit the low scores. Overall, the score guideline has a real impact on the VLM score, allowing to guide it and improve the correlation with the human evaluator.

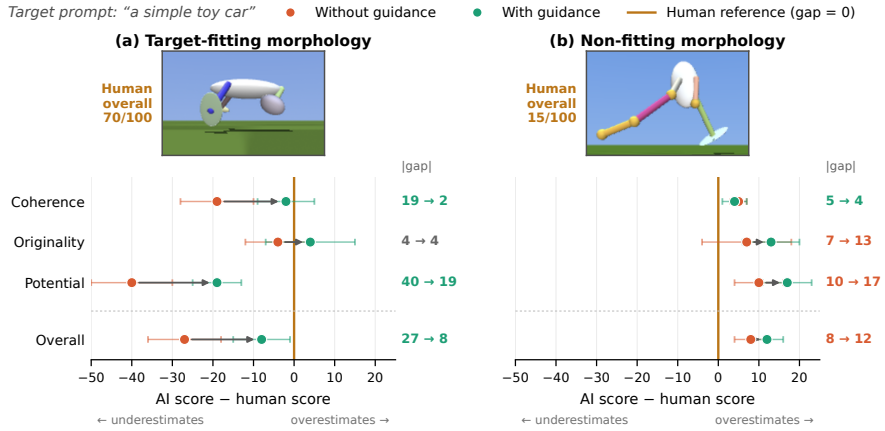


Fig. 8: **Impact of score range guidance on VLM scoring**. We compare VLM scores obtained without guidance and with guidance against human scores, for two morphologies: a target-fitting one (70/100 by humans) and a non-fitting one (15/100 by humans) .

Consistency

Consistent scoring is critical for a stable evolutionary process, as high variance would introduce noise into fitness selection.

Figure 9 reports morphology consistency: the VLM was asked 20 times to score the same average-looking car and a random morphology. The standard deviation remained around 10% of the mean in both cases, and the good morphology was correctly ranked above the random one in all 20 trials, confirming reliable and consistent image scoring.

Figure 10 reports the equivalent experiment for video-based behavior evaluation, targeting "a robot jumping as high as possible". Results are markedly different: the standard deviation reached 50% of the mean. Based on this sample, **the target-fitting video would be selected correctly only $\tilde{6}7\%$ of the time**. Averaging 3 independent VLM score per individual raises this to $\tilde{7}8\%$, and this is therefore applied throughout all behavior search experiments. Without this correction, scoring noise would consistently mislead the evolutionary process toward non-meaningful solutions.

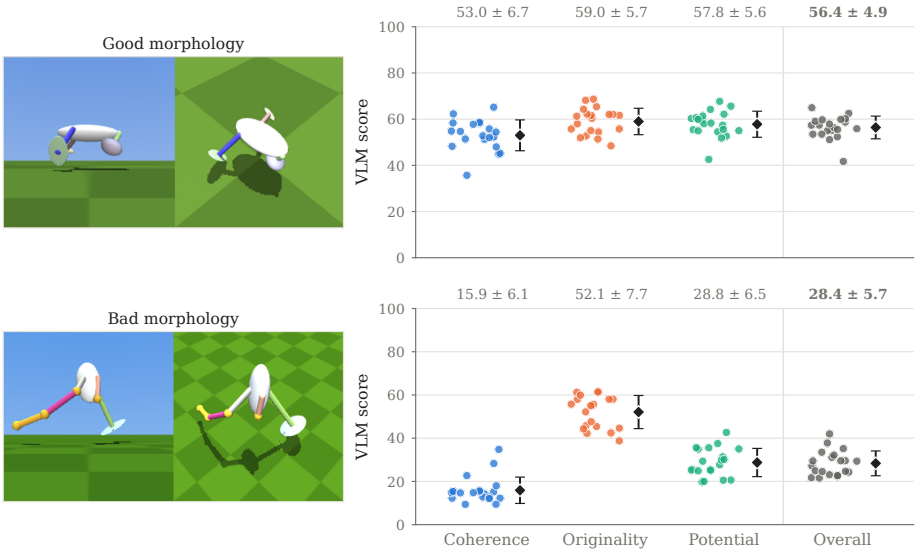


Fig. 9: **VLM morphology scoring consistency across 20 evaluations** : Mean and standard deviation for a "*Car*" morphology target. VLM scores are repeatable for both a "good" and a "bad" morphology example, with a standard deviation remaining below 10% of the mean in both cases.

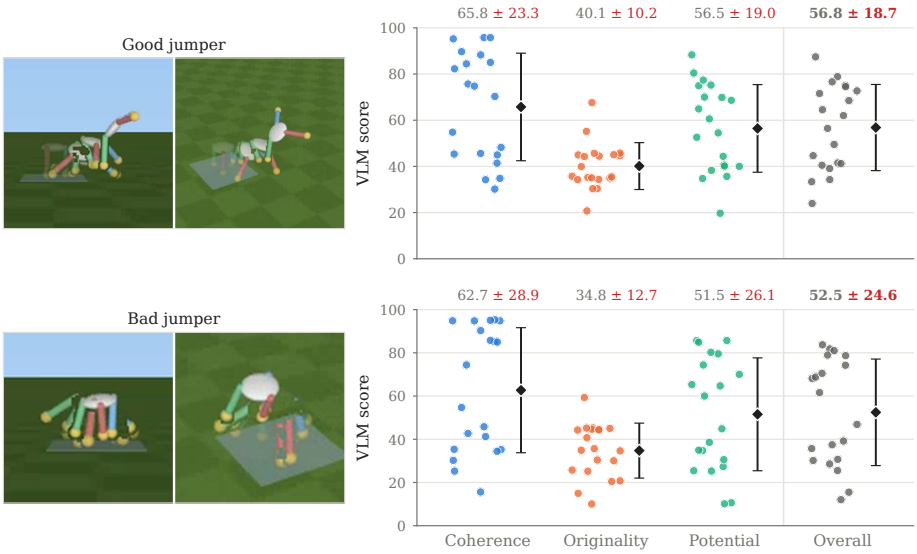


Fig. 10: **VLM behavior scoring consistency across 20 evaluations** : Mean and standard deviation for a "*Jumping as high as possible*" behavior target. VLM scores are not repeatable for both a "good" and a "bad" morphology example, with a standard deviation reaching 50% of the mean in both cases.

These experiments confirm that VLM-based scoring is reliable and human-aligned for image-based morphology evaluation, but significantly less for video-based behavior scoring, where low consistency, spatial hallucinations, and sensitivity to prompt remain the main obstacles.

4.2 Morphology Search

We ran multiple simulations targeting different objectives such as "kangaroos", "crabs", or "lamps". In all cases, the MAP-Elites strategy, thanks to its diversity search and archiving method, found better or at least equivalent solutions compared to the basic $(\mu + \lambda)$ approach. Indeed, the $(\mu + \lambda)$ strategy leads very quickly to approximately target-fitting morphologies (figure 11), but often fall into local optimum, which is a consequence of the basic strategy approach. MAP-Elites is able to avoid most of these issues by keeping a wide and diverse population of candidate (figure 12).

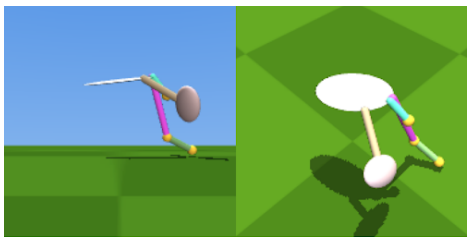


Fig. 11: $(\mu + \lambda)$ strategy for "*ergonomic and stylish lamp*" leading to a small lamp with a two-segment articulated base, a circular lampshade, and a handle for position adjustment. A coherent match to the target description for a modest computational budget: 40 init individuals, 20 offspring and 5 best performers kept for all 10 generations.

The MAP-Elites approach for simple morphology search is more conclusive. It effectively found great solutions, and diverse ones : most of the MAP-Elite cells are filled with different solutions. The VLM is clearly able to assign descriptors to morphologies, classifying them based on their subjective types, concept and characteristics. This confirms the ability of the VLM to give subjective judgments on visual morphologies.

To verify the results, we ran the simulation with other targets, for example we looked for tree-like morphologies (available on the website).

4.3 Controller Search

The results for dynamic behavior search are not as good as for morphology search. The VLM struggled to distinguish great results from random movements due to **frequent spatial hallucinations**, which is the main limitation.



Fig. 12: MAP-Elites strategy applied to the target "*ergonomic and stylish lamp*". This strategy produces highly diverse solutions, with the VLM autonomously assigning subjective lamp-type labels (reading lamp, deco lamp, spotlight, bedside lamp, etc.) as behavioral descriptors. The archive is structured as a 4×7 grid of feature niches along two axes: adjustability and design. The grid is only partially filled due to the limited computational budget, kept equivalent to the $(\mu + \lambda)$ experiment (Figure 11): 40 initial random individuals, with 20 mutated offspring across 10 generations.

Across multiple experiments with different targets, we found that the VLM is unable to capture behavioral details, only the outline.

All descriptors related to general shapes and movements are well handled. The difficulty arises when it comes to finer sub-movements or small details and nuances, due to the VLM's awareness resolution being too low to capture details in abstract videos where spatial reasoning is crucial. This is very likely due to poor video frame rate (1 frame per seconds), which is the default settings of the model used. Increasing the frame rate may improve the performance, but it would significantly extend the inference times (already long with $\tilde{6}$s per individual at 1fps of 5s videos), which is not viable for optimization purposes.

More experimental details, results, and video examples are available on the accompanying website in the behavior section.

5 Further Work

This exploratory work opens many opportunities for pursuing and exploring different aspects.

Many experiments in this paper were limited by the virtual environment rendering. Repeating the experiments in a more realistic environment may drastically improve the performance of the VLM as a reward model and reduce spatial hallucinations, thanks to richer context and greater proximity to the model's training domain.

All benchmark experiments should be updated with local models to avoid the media upload cost, or newer models, such as Gemini 3.5 Flash. For example, we evaluated the correlation 4.1 with it and found a very similar result, but it may not be the case for other benchmark.

It would be very interesting to try body-brain dual evolution guided by a VLM as a reward model.

Another aspect that would be extremely interesting is to exploit more of the VLM's capabilities: not only using it as a reward model, but also as an archive manager by dynamically changing the descriptors, or an evolution driver by guiding the mutation.

6 Conclusion

The objective of the study is establish whether a zero-shot VLM is able to reproduce the "subjective" human way of scoring for searching "interesting" design. We ends up with these key-takeaway :

- Image-based zero-shot VLM grading reliably guides morphology evolution toward different subjective target-fitting shapes, achieving a Pearson correlation of 0.5 with human evaluators when using score range guidance.
- The VLM successfully assigns abstract language-based feature descriptors such as concepts or adjectives, as long as classification relies on basic shape/outline comparisons.
- Video-based scoring captures the outline motion, but struggles because of spatial hallucinations, high scoring inconsistency, and low frame rate hiding the details.
- Spatial reasoning, output consistency, and inference speed are the main bottlenecks, all of which are improving rapidly with each new model generation, suggesting strong potential for VLM-based reward models in the near future.

This paper is a proof of concept focusing on virtual robotic design and subjective targets, with a zero-shot VLM reproducing a human evaluator. It complements other work on FMs being used as flexible reward models or alternative component for morphology and behavior search. The approach may be expanded to many other domains where visual analysis is relevant, such as creativity or visual computing exploration.

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